Lab M10.04 – Choosing the Right File Format (Application Lab)

Scenario 1: Streaming Clickstream Ingestion

Scenario 2: Daily Sales Analytics

Scenario 3: Data Exchange with External Partners

Scenario 4: Ad-Hoc Analyst Exploration

Scenario 5: Long-Term Archival Storage

Step 1: Scenario Analysis & Decisions

<u>Scenario 1: Streaming Clickstream Ingestion</u>

Selected Format: JSON

Write pattern: High-throughput streaming. JSON supports append-heavy, event-based ingestion easily.

Schema: flexibility (new fields can appear without breaking ingestion).

Immediate availability: No heavy encoding step needed.

Human-readable: Critical for debugging live events.

Trade-offs:

- Poor compression vs columnar formats, therefore very low storage efficiency
- Low read efficiency, poor analytics

<u>Scenario 2: Daily Sales Analytics</u>

Selected Format: Parquet

Read pattern: Analytical queries with column pruning.

Performance: Reads only required columns, huge speed gains.

Compression: Very efficient, reduces storage cost.

Batch processing: Ideal for once-per-day writes.

Trade-offs:

- Not human-readable
- Slower writes than JSON/CSV

Scenario 3: Data Exchange with External Partners

Selected Format: CSV

Interoperability: Universally supported (Excel, databases, scripts).

Human-readable: Easy inspection.

High write efficiency

Trade-offs:

- No schema enforcement
- No compression, low storage efficiency
- Inefficient for complex analytics

Scenario 4: Ad-Hoc Analyst Exploration

Selected Format: JSON

Schema flexibility.

Exploration: Nested/semi-structured data is easier to inspect.

Human-readable: Helps understanding raw data quickly.

Trade-offs:

- Slower queries than Parquet
- Low storage efficiency

Scenario 5: Long-Term Archival Storage

Selected Format: Parquet

Low storage cost.

Write efficiency is not needed.

Access pattern: optimized for efficient analytics.

Trade-offs:

- Not human-readable

Step 2: Decision Table

Scenario	Format	Primary Justification	Read Pattern	Write Pattern	Schema	Performance	Interoperability	Trade-offs
Streaming Clickstream	JSON	Flexible, fast streaming writes	Exploratory	Continuous streaming	Evolving	Fast writes, slow reads	High	Poor compression, costly analytics
Daily Sales Analytics	Parquet	Columnar analytics efficiency	Aggregations, filtering	Daily batch	Stable	Excellent read and high compression	Medium	Not human-readable
Data Exchange	CSV	Interoprability	Full-row reads	Batch / moderate	Stable	Moderate	Very high	No schema enforcement
Ad-Hoc Exploration	JSON	Flexible and inspectable	Unpredictable	Small-medium	Evolving	Moderate	High	Slower queries
Long-Term Archive	Parquet	Best compression for storage	Rare analytical	Append-only	Stable	High compression	Medium	Not human-readable

Step 3: Justification Details

<u>Scenario 1: Streaming Clickstream Ingestion</u>

Format: JSON

Read/write: Optimized for high-ingest streaming.

Schema evolution: flexible.

Performance: Fast ingestion, slower downstream queries.

Interoperability: Works across all ingestion systems.

<u>Scenario 2: Daily Sales Analytics</u>

Format: Parquet

Read/write: Batch writes and heavy analytical reads.

Schema enforcement.

Performance: Column pruning and predicate pushdown (high read efficiency).

Interoperability: Widely supported in modern data stacks.

Trade-off: Debugging harder due to binary format.

Not human- readable.

Scenario 3: Data Exchange

Format: CSV

Read/write: Simple row-based reads/writes.

Schema: Implicit (header-based), fine for stable schema but no schema enforcement.

Performance: Acceptable for low to moderate data sizes.

Interoperability: Maximum compatibility.

Scenario 4: Ad-Hoc Exploration

Format: JSON

Read/write: Flexible for partial reads and sampling.

Schema: Ideal for unknown or evolving structures.

Performance: Acceptable for small-medium datasets.

Interoperability: Widely supported.

Not efficient for scaling analytics.

Scenario 5: Long-Term Archive

Format: Parquet

Read/write: Optimized for append.

Schema: Strong schema enforcement

Performance: Excellent compression (storage) and efficient analytics.

Interoperability: Widely supported in modern data stacks.

Not human- readable.

Step 4: Reusable Decision Framework

Format Selection Decision Tree

1. Primary Access Pattern

    Analytical → Parquet / ORC

    Transactional / row-based → CSV / JSON

    Exploratory → JSON (or CSV if simple)

2. Write Pattern

Streaming → JSON

Batch → Parquet / ORC

Append-only → Parquet / ORC

3. Schema Evolution

Needed → JSON, Parquet

Stable → CSV, Parquet, ORC

4. Interoperability

Universal → CSV, JSON

Controlled ecosystem → Parquet, ORC

5. Primary Concern

Storage cost → ORC, Parquet

Query performance → Parquet, ORC

Human readability → CSV, JSON

Flexibility → JSON